TED (15)-6071

REVISION-2015

MODEL QUESTION PAPER

SIXTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

INSTRUMENTATION AND PROCESS CONTROL

Maximum marks: 100 | Time: 3 hrs

Marks

PART-A

(Maximum marks: 10)

Answer the following questions in one or two sentences. Each question carries two marks

I	1)	List any two flow measuring instruments used in process industries	2
	2)	Define Peltier effect.	2
	3)	Define dew point	2
	4)	Identify any two applications of atomic absorption spectroscopy	2
	5)	Define dead time	2

PART-B

(Maximum marks: 30)

II		Answer any five of the following question. Each question carries 6 marks	
	1)	Summarize the characteristics of a measuring instrument.	6
	2)	Illustrate the working of a gas flow meter.	6
	3)	Summarize the sensing devices of a pH meter	6
	4)	Illustrate the working of an Oswald viscometer.	6
	5)	Summarize various methods of composition analysis used in process industries	6
	6)	State the advantages and disadvantages of distributed control system (DCS).	6
	7)	Sketch the standard instrumentation symbols for a) flow control valve b) solenoid valve c) pressure relief valve and d) locally mounted instrument	6

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each question carries 15 marks)

Module-I

III	a)	Demonstrate the working of a pressure spring thermometer	7
	b)	Illustrate the working of electronic signaling type tank level measurements.	8
		OR	
IV	a)	Illustrate the working of RTD.	8
	b)	Discuss the working of bellow gauge used in industrial pressure measurements.	7

Module-II

V	a)	Select a suitable instrument for continuous specific gravity measurement and illustrate it's working.	7
	b)	Illustrate the method of humidity measurement with the help of a wet and dry bulb thermometer.	8
		OR	
VI	a)	Illustrate the method of humidity measurement using dew point recorder.	7
	b)	Illustrate the working of reference electrode in a pH meter	8

Module-III

VII	a)	Suggest a suitable method for analyzing sodium content in steam and illustrate its working principle.	8
	b)	Summarize the advantages of chromatography in composition analysis.	7
		OR	
VIII	a)	Explain the working of a mass spectrometer.	8
	b)	Discuss in detail, the composition analysis using atomic adsorption spectroscopy.	7

Module-IV

IX	a)	Prepare a piping and instrumentation diagram (P&ID) to show the major control schemes in a single effect evaporator	8
	b)	Summarize the forces acting on the output member of a final control element.	7
		OR	
X	a)	Discuss the functions of an A/D converter and D/A converter	8
	b)	Summarize the major components of Supervisory control and data acquisition system (SCADA) .	7